

100%

Excellent!

the material from Chapters 13, 14, 15 of OpenStax College Physics. Please write your name
read each question. Credit is only awarded if you answer correctly and clearly show all
credit is not awarded for missing work, unclear work, or the use of memorized equations
may be awarded for insufficient work that suggests some degree of understanding. Half credit
is awarded for answers with incorrect or missing units.

1. Find the increase in entropy as 5.3 kg of ice, originally at 0.0°C, melts to form water at 0.0°C at a particular pressure. The latent heat of fusion for water is 334 kJ/kg.

ANSWER: 6.48 kJ/K ✓

$$\begin{aligned}\Delta S &= \frac{Q}{T} = \frac{m L_f}{T} \\ &= \frac{(5.3 \text{ kg})(334 \text{ kJ/kg})}{0 + 273.15 \text{ K}} \\ &= 6.48 \text{ kJ/K}\end{aligned}$$

2. Charles is a design engineer for a lightbulb company. He manufactures a gas-filled incandescent light bulb so that the gas inside the bulb is at atmospheric pressure when the bulb has a temperature of 7°C. What is the pressure in the light bulb when it rises to 238°C? The coefficient of volume expansion for the glass light bulb is $30 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$.

ANSWER: $1.84 \times 10^5 \text{ Pa}$ ✓

$$P_1 = 101325 \text{ Pa}$$

$$T_0 = 7^\circ\text{C}$$

$$T_f = 238^\circ\text{C}$$

$$\beta = 30 \times 10^{-6} \frac{1}{^\circ\text{C}}$$

$$P_2 = ?$$

$$\frac{P_1 V_i}{T_0} = \frac{P_2 V_f}{T_f}$$

$$\Delta V = V_f - V_i$$

$$\Delta V = \beta V_i \Delta T$$

$$V_f - V_i = \beta V_i \Delta T$$

$$V_f = \beta V_i \Delta T + V_i$$

$$V_f = V_i (1 + \beta(T_f - T_i))$$

$$\frac{P_1 V_i}{T_0} = \frac{P_2 V_f}{T_f}$$

$$\frac{P_1 V_i T_f}{T_0 V_f} = P_2$$

$$\frac{P_1 V_i T_f}{T_0 (1 + \beta(T_f - T_i))} = P_2$$

$$\frac{P_1 T_f}{T_0 (1 + \beta(T_f - T_i))} = P_2$$

$$\frac{(101325 \text{ Pa})(511.15 \text{ K})}{(280.15 \text{ K})(1 + (30 \times 10^{-6})(231))} = P_2$$

$$P_2 = 1.84 \times 10^5 \text{ Pa}$$

3. Patricia's 75-L (19.8-gal) steel gasoline tank is full of gas. Both the tank and the gasoline have an initial temperature of 12°C. How much gasoline has leaked out due to thermal expansion by the time both the tank and the gasoline have warmed to 30.1°C? The coefficient of volume expansion for steel is $35 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$ and for gasoline it is $950 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$. Show all work.

ANSWER: 1.98 L ✓

$$\Delta V_s = \beta_s V \Delta T$$

$$\Delta V_g = \beta_g V \Delta T$$

$$V^* = \Delta V_g - \Delta V_s$$

$$V^* = \beta_g V \Delta T - \beta_s V \Delta T$$

$$V^* = V \Delta T (\beta_g - \beta_s)$$

$$= (75)(28.9)(950 \times 10^{-6} - 35 \times 10^{-6})$$

$$= 1.98 \text{ L}$$

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4. What is the average kinetic energy of a gas molecule at 30°C?

ANSWER: $6.28 \times 10^{-21} \text{ J}$

$$\begin{aligned}\overline{KE} &= \frac{3}{2} k_B T \\ &= \frac{3}{2} (1.38 \times 10^{-23} \frac{\text{J}}{\text{K}}) (303.15 \text{ K}) \\ &= 6.28 \times 10^{-21} \text{ J}\end{aligned}$$

5. Calculate the work output of a Carnot engine operating between temperatures of 549 K and 329 K for 7630 J of heat transfer to the engine.

ANSWER: 3057.6 J

$$\eta_c = 1 - \frac{T_c}{T_h}$$

$$\text{since } \frac{T_c}{T_h} = \frac{Q_c}{Q_h}$$

$$J(1 - \frac{K}{K}) = J$$

$$\Rightarrow 1 - \frac{T_c}{T_h} = \frac{W}{Q_h}$$

$$W = Q_h (1 - \frac{T_c}{T_h}) = (7630)(1 - \frac{329}{549}) = 3057.6 \text{ J}$$

6. Julia puts a 0.39-kg aluminum pot on the stove and fills it with 0.61 liters of water. Both the pot and the water are initially at 6.8°C. How much heat is required to bring them both to 60.4°C? The specific heat value of water is 4186 J/kg°C and for aluminum it is 900 J/kg°C.

ANSWER: 155679.1 J

$$m_{Al} = 0.39 \text{ kg}$$

$$V_{H_2O} = 0.61 \text{ L}$$

$$T_0 = 6.8^\circ\text{C}$$

$$T_f = 60.4^\circ\text{C}$$

$$C_{H_2O} = 4186 \text{ J/kg}^\circ\text{C}$$

$$C_{Al} = 900 \text{ J/kg}^\circ\text{C}$$

$$\Sigma Q = Q_{Al} + Q_{H_2O}$$

$$= m_{Al} C_{Al} \Delta T + m_{H_2O} C_{H_2O} \Delta T$$

$$= \Delta T (m_{Al} C_{Al} + m_{H_2O} C_{H_2O})$$

$$= (53.6)((0.39)(900) + (1)(0.61)(4186))$$

$$= 155679.1 \text{ J}$$

$$m_{H_2O} = dV$$

$$d_{H_2O} = 1 \text{ kg/L}$$

+3

7. Suppose that there is heat transfer of 54 J to a system, while the system does 30 J of work. Later, there is heat transfer of 42 J out of the system while 42 J of work is done on the system. What is the net change in internal energy of the system?

ANSWER: 24 J

$$Q_1 = 54 \text{ J} \quad W_1 = 30 \text{ J}$$

$$Q_2 = -42 \text{ J} \quad W_2 = -42 \text{ J}$$

$$\Delta U = \sum Q - \sum W$$

$$= (54 + (-42)) - (30 + (-42))$$

$$= 12 - (-12)$$

$$= 24 \text{ J}$$

8. Robert inflates his bicycle tire to an absolute pressure of 0.77 MPa (a gauge pressure of about 96 psi) at a temperature of 11.7°C. What is the tire's gauge pressure after the temperature has risen to 34.8°C? Assume there are no leaks and the volume of the tire does not change.

ANSWER: 731118.4 Pa

$$P_1 = 0.77 \times 10^6 \text{ Pa}$$

$$T_0 = 11.7^\circ \text{C}$$

$$T_f = 34.8^\circ \text{C}$$

$$\frac{P_1 V_1}{n_1 T_0} = \frac{P_2 V_2}{n_2 T_f}$$

$$\frac{P_1}{T_0} = \frac{P_2}{T_f}$$

$$\frac{P_1}{T_0} = \frac{P_g + P_{atm}}{T_f}$$

$$\frac{P_1 T_f}{T_0} - P_{atm} = P_g$$

$$P_g = \frac{(0.77 \times 10^6)(34.8 + 273.15)}{(11.7 + 273.15)} - 101325$$

9. Natalie is producing heat at a rate of about 258 W while doing light exercise. At what rate must water evaporate from her body to get rid of the produced energy? The latent heat of vaporization of water is 2430 kJ/kg at body temperature, 37°C.

ANSWER: 6.37 g/min

$$P = \dot{Q} = 258 \text{ W}$$

$$Q = m L_v$$

$$\frac{Q}{t} = \frac{m L_v}{t}$$

$$\dot{Q} = \dot{m} L_v$$

$$\frac{258}{2430 \times 10^3} = \dot{m}$$

$$\frac{\text{J/s}}{\text{J/kg}} \rightarrow \text{kg/s}$$

$$\dot{m} = 6.37 \text{ g/min} \quad \frac{\text{kg}}{\text{s}} \left| \frac{1000 \text{ g}}{1 \text{ kg}} \right| \frac{60 \text{ s}}{1 \text{ min}}$$

10. Stephen wants to use his pressure cooker to cook food in liquid water at a temperature of 152°C . To determine the pressure required inside the pressure cooker for water to boil at this temperature, he can use the Clausius-Clapeyron equation for water:
- $$P(T) = (1 \text{ kPa}) \cdot \exp\left[\left(\frac{1}{273.15 \text{ K}} - \frac{1}{T}\right)\right]$$
- In this equation, T is the temperature in Kelvin and P is the pressure in kPa. Given that the lid of the pressure cooker is a disk with a diameter of 40 cm, what is the force that the lid must be able to withstand at the required pressure?

ANSWER: $50.26 \text{ kPa} \cdot \text{m}^2$ or kN

$$P_g = \frac{F}{A}$$

$$F = (P - P_{\text{atm}}) \cdot A$$

$$= (P(T) - P_{\text{atm}}) (\pi r^2)$$

$$P(T) = (1 \text{ kPa}) e^{\left(\frac{1}{273.15 \text{ K}} - \frac{1}{425.15 \text{ K}}\right)}$$

$$= 501.2902471 \text{ kPa}$$

$$P_{\text{atm}} = 101325 \text{ Pa} = 101.325 \text{ kPa}$$

$$d = 40 \times 10^{-2} \text{ m}$$

$$r = 0.2 \text{ m}$$

$$r^2 = 0.04 \text{ m}^2$$

$$F = (P(T) - P_{\text{atm}}) (\pi r^2)$$

$$F = 50.26 \text{ kPa} \cdot \text{m}^2$$

11. Aaron drives a 3100-kg truck and descends a 27-m-tall hill at a constant speed. What is the final temperature of 300 kg of brake material if its temperature is 8°C at the top of the hill? Assume the specific heat of the brake material is $710 \text{ J/kg}^\circ\text{C}$. Assume the brakes do all the work to maintain the speed.

ANSWER: 11.9°C

$$m_t = 3100 \text{ kg}$$

$$\Delta h = 27 \text{ m}$$

$$\Delta V = 0$$

$$T_o = 8^\circ\text{C}$$

$$m_b = 300 \text{ kg}$$

$$c_b = 710 \text{ J/kg}^\circ\text{C}$$

$$\Delta E_{\text{tot}} = (KE_f + PE_f) - (KE_i + PE_i)$$

$$= mg \Delta h$$

$$E_{\text{tot}} = W_b$$

$$\Delta V = 0 \rightarrow w = Q$$

$$m_t g \Delta h = m_b c_b (T_f - T_i)$$

$$\frac{m_t g \Delta h}{m_b c_b} + T_i = T_f$$

$$\frac{\text{kg} \cdot \frac{\text{m}}{\text{s}^2} \cdot \text{m}}{\frac{\text{kg}}{\text{s}^2} \cdot \frac{\text{J}}{\text{kg}^\circ\text{C}}} + ^\circ\text{C}$$

$$\frac{\text{kg} \cdot \frac{\text{m}^2}{\text{s}^2}}{\frac{\text{J}}{\text{s}^2}} + ^\circ\text{C}$$

$$\frac{\text{J}}{\text{s}^2} \quad \frac{\text{kg} \cdot \frac{\text{m}^2}{\text{s}^2}}{\text{J}}$$

12. Chloe's 64-L (16.9-gal) steel gasoline tank is full of gas. Both the tank and the gasoline have an initial temperature of 12.7°C . After both the tank and the gasoline warm to 37.3°C , what is the pressure in the gasoline tank if the gasoline cannot spill out? The coefficient of volume expansion for steel is $35 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$ and for gasoline it is $950 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$. Assume the bulk modulus B for gasoline is $1.00 \times 10^9 \text{ N/m}^2$. Show all work.

ANSWER: $2.25 \times 10^7 \text{ N/m}^2$

$$\Delta V_s = \beta_s V \Delta T$$

$$\Delta V_g = \beta_g V \Delta T$$

$$V^* = \beta_g V \Delta T - \beta_s V \Delta T$$

$$= V \Delta T (\beta_g - \beta_s)$$

$$P = \beta \Delta T (\beta_g - \beta_s)$$

$$= (1 \times 10^9 \frac{\text{N}}{\text{m}^2}) (24.6\%) (9.15 \times 10^{-4} \frac{1}{^\circ\text{C}})$$

$$= 2.25 \times 10^7 \frac{\text{N}}{\text{m}^2}$$

x3

PHY201 Exam on chapter(s): 13, 14, 15

1. 6.48 kJ/K
2. 183.601 kPa
3. 1.98 L
4. 6.28×10^{-21} J
5. 3057.56 J
6. 155.68 kJ
7. 24 J
8. 731.12 kPa
9. 6.37 g/min
10. 50.26 kN
11. 11.85 °C
12. 22.51 MPa